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SQL

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Clear

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Run

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select * from drivers;

▼ 857 rows ×

DRIVERID	DRIVERREF	CODE	FORENAME	SURNAME	DOB	NATIONALITY
0	hamilton	HAM	Lewis	Hamilton	1985-01-07 00:00:00	British
1	heidfeld	HEI	Nick	Heidfeld	1977-05-10 00:00:00	German
2	rosberg	ROS	Nico	Rosberg	1985-06-27 00:00:00	German
3	alonso	ALO	Fernando	Alonso	1981-07-29 00:00:00	Spanish
4	kovalainen	KOV	Heikki	Kovalainen	1981-10-19 00:00:00	Finnish
5	nakajima	NAK	Kazuki	Nakajima	1985-01-11 00:00:00	Japanese
6	bourdais	BOU	Sébastien	Bourdais	1979-02-28 00:00:00	French
7	raikkonen	RAI	Kimi	Räikkönen	1979-10-17 00:00:00	Finnish
8	kubica	KUB	Robert	Kubica	1984-12-07 00:00:00	Polish
9	glock	GLO	Timo	Glock	1982-03-18 00:00:00	German
10	sato	SAT	Takuma	Sato	1977-01-28 00:00:00	Japanese
11	piquet_jr	PIQ	Nelson	Piquet Jr.	1985-07-25 00:00:00	Brazilian
12	massa	MAS	Felipe	Massa	1981-04-25 00:00:00	Brazilian
13	coulthard	COU	David	Coulthard	1971-03-27 00:00:00	British
14	trulli	TRU	Jarno	Trulli	1974-07-13 00:00:00	Italian
15	sutil	SUT	Adrian	Sutil	1983-01-11 00:00:00	German
16	webber	WEB	Mark	Webber	1976-08-27 00:00:00	Australian
17	button	BUT	Jenson	Button	1980-01-19 00:00:00	British
18	davidson	DAV	Anthony	Davidson	1979-04-18 00:00:00	British
19	vettel	VET	Sebastian	Vettel	1987-07-03 00:00:00	German

857 rows

Prev1 / 43Next

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Explain

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PREDICT COUNT(qualifying.*, 0, 365, days)
FOR EACH drivers.driverId

▼ 857 rows ×

ENTITY	ANCHOR_TIMESTAMP	TARGET_PRED
0	2009-10-31T11:00:00	18
1	2009-10-31T11:00:00	18
2	2009-10-31T11:00:00	18
3	2009-10-31T11:00:00	18
4	2009-10-31T11:00:00	18
5	2009-10-31T11:00:00	18
6	2009-10-31T11:00:00	18
7	2009-10-31T11:00:00	18
8	2009-10-31T11:00:00	18
9	2009-10-31T11:00:00	18
10	2009-10-31T11:00:00	0
11	2009-10-31T11:00:00	18
12	2009-10-31T11:00:00	18
13	2009-10-31T11:00:00	18
14	2009-10-31T11:00:00	18
15	2009-10-31T11:00:00	18
16	2009-10-31T11:00:00	18
17	2009-10-31T11:00:00	18
18	2009-10-31T11:00:00	0
19	2009-10-31T11:00:00	18

857 rows

Prev1 / 43Next

♀ EXPLANATION

The model predicts that driverId 0 (a British driver with code "HAM") will have 18 qualifying appearances in the next 365 days.

Key Factors Behind This Prediction:

1. STRONG SEASON PERFORMANCE (STANDINGS POINTS AND WINS)

- From the global cohort analysis, higher points in the standings correlate with more qualifying appearances:
 - Drivers with 5-8 points average about 1.38 qualifying appearances.
 - Drivers with 7-9+ points and beyond reach up to 3.23 to 3.49 appearances.
- This specific driver shows multiple standings records with increasing points over time (e.g., points ranging from 14 to 98) and several race wins increasing from 1 up to 5.
- While individual standings cells have zero attribution in this subgraph, the driver's consistent high points and multiple wins imply strong performance globally, which drives the prediction for a high count of qualifying starts.

2. RECENT RACING ACTIVITY

- The driver's results show regular race participations with finishing positions mostly within competitive ranges (e.g., ranks between 1 and 19), indicating active and successful engagement.
- This consistent participation in races links to steady qualifying attempts.
- The model likely infers that a driver frequently racing with good ranks and points will also have a higher number of qualifying sessions.

3. QUALIFYING POSITION GLOBAL TRENDS

- Cohort analysis shows drivers qualifying in top positions have notably higher qualifying attempt rates:
 - Qualifying positions between 1-7 correspond to an average target of over 9 qualifying appearances.
 - This driver has qualifying positions clustered mostly in the 1-5 range (positions like 1, 3, 4, 5), aligning with cohorts where drivers are predicted to qualify frequently.
- The subgraph nodes for qualifying positions have zero local attribution scores, suggesting the model uses this global pattern rather than emphasizing any particular qualifying event for this driver.

4. ATTRIBUTES WITH LOW LOCAL SCORES

- Interestingly, all node cell scores are zero for every individual qualifying, standings, and results attribute. This implies the model relies more on aggregated patterns and cohort statistics for prediction than on any single event's data.
- The driver's nationality and code also have zero scores, indicating those features do not directly drive the prediction.

Summary Insight:

This British driver is predicted to qualify approximately 18 times in the next year due to a history of strong points accumulation, multiple wins, and consistently high qualifying positions in recent seasons. Although no single qualifying event carries a critical influence, the model's understanding from global cohort trends is that drivers performing this well usually appear frequently in qualifying rounds. Hence, the model confidently predicts a high volume of qualifying sessions for this driver.

⏏ MODEL PERFORMANCE

10.7 %

MAE

96.6%

MSE

98.3%

RMSE

⋮

PQL

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Explain

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Run

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PREDICT AVG(results.positionOrder, 0, 30, days)
FOR EACH drivers.driverId

▼ 857 rows ×

ENTITY	ANCHOR_TIMESTAMP	TARGET_PRED
0	2009-11-01T11:00:00	9
1	2009-11-01T11:00:00	10.7958860397
2	2009-11-01T11:00:00	8.5
3	2009-11-01T11:00:00	10.5
4	2009-11-01T11:00:00	10
5	2009-11-01T11:00:00	12.2430734634
6	2009-11-01T11:00:00	13.5
7	2009-11-01T11:00:00	8.8709630966
8	2009-11-01T11:00:00	11
9	2009-11-01T11:00:00	9
10	2009-11-01T11:00:00	11
11	2009-11-01T11:00:00	13
12	2009-11-01T11:00:00	10.5
13	2009-11-01T11:00:00	8.5
14	2009-11-01T11:00:00	10.8967552185
15	2009-11-01T11:00:00	15
16	2009-11-01T11:00:00	8.2761077881
17	2009-11-01T11:00:00	6.5
18	2009-11-01T11:00:00	14
19	2009-11-01T11:00:00	6.5

857 rows

Prev1 / 43Next

♀ EXPLANATION

The model predicted the driver's average positionOrder as 16.5.

From the global cohort analysis for the column `results.positionOrder` (hop=1), we observe the following:

- The population distribution is fairly uniform across positionOrder cohorts from 5 to 15+.
- The average targets (i.e., typical positionOrder) gradually rise from about 9.7 in the 2 to 7 range up to 12-15 in the higher positionOrder cohorts.
- The cohort "[13.59 - 16]" has a target average of 15.27, close to the predicted 16.5, indicating the model expects a finishing position in the mid-to-late teens for this driver.

Looking at the local subgraph explanation for this driver (driverId=152, nationality Spanish with code ALG):

- All attribute attribution scores are zero, indicating no particular data points strongly push the prediction up or down here.
- The driver's results nodes show positionOrder values mostly clustered around 14 to 20, e.g., race results with positionOrder values of 14, 15, 16, 18, and 20.
- These actual race finishes align well with the predicted average finish (16.5).
- The driver has no wins or points in the associated standings and consistently low standings positions (from 21st to 24th) in recent races during 2009.
- The driver races with a constructor of Italian nationality, which globally associates with middling average positions (~12.1 target).

In summary, the model predicts an average finishing position of about 16.5 for this driver because their recent race results predominantly show finishes in the mid-to-high teens (14 to 20), matching the trends from global cohorts where similar positionOrder values correspond to similar expected averages. The absence of standout positive signals like wins, points, or very good qualifying results (all scores 0) suggests a generally stable but not competitive performance level, which results in the mid-late teen average finishing position predicted.

⏏ MODEL PERFORMANCE

2.690

MAE

14.233

MSE

3.773

RMSE

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PQL

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Explain

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Evaluate

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Run

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PREDICT AVG(results.milliseconds, 0, 30, days)
FOR EACH constructors.constructorId

▼ 211 rows ×

ENTITY	ANCHOR_TIMESTAMP	TARGET_PRED
0	2009-11-01T11:00:00	5515288.5
1	2009-11-01T11:00:00	5522857
2	2009-11-01T11:00:00	5552958.5
3	2009-11-01T11:00:00	5558006.5
4	2009-11-01T11:00:00	5644739.5
5	2009-11-01T11:00:00	5522153.5
6	2009-11-01T11:00:00	5566026.5
7	2009-11-01T11:00:00	5896700
8	2009-11-01T11:00:00	5553494.5
9	2009-11-01T11:00:00	5720181.5
10	2009-11-01T11:00:00	5652764.5
11	2009-11-01T11:00:00	5844520.5
12	2009-11-01T11:00:00	5759149
13	2009-11-01T11:00:00	5750715
14	2009-11-01T11:00:00	5600729
15	2009-11-01T11:00:00	5603198
16	2009-11-01T11:00:00	5618837
17	2009-11-01T11:00:00	5659581
18	2009-11-01T11:00:00	5600090.5
19	2009-11-01T11:00:00	5629073

211 rows

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♀ EXPLANATION

The model predicts that the average race completion time in milliseconds for constructor #7 (Japanese nationality) is approximately 5,897,606 ms.

Looking at the global column analysis across many constructors and races:

- The number of constructor results (races participated in) is highly predictive. Constructors with 0-22 race results have a higher average milliseconds time (~5,786,526 ms) compared to those with over 32 race results (~5,654,134 ms). This implies that more experienced constructors tend to finish races faster.
- Points scored in constructor results have relatively small variance in predicted milliseconds (all cohorts near ~5.6 million ms), suggesting points alone weakly affect race time prediction.
- Race dates clustered into cohorts show competitor times average around 5.66 to 5.69 million ms, reflecting temporal consistency in race durations.
- Nationality of constructors presents some variance: Japanese constructors have a predicted average milliseconds of ~5,663,644 across all data, relatively close to the overall target mean (~5.66 million ms).

In the local subgraph for constructor #7:

- All individual race results linked to this constructor report zero points repeatedly and the race dates range mostly from 2006 to 2008.
- The model does not assign high attribution scores to any particular cell within constructor results, qualifying, races, or constructor standings for constructor #7 — all feature importance scores are zero or negligible.
- Notably, the constructor's nationality "Japanese" also has a zero influence score, indicating it neither strongly raises nor lowers the predicted average milliseconds.

In summary:

- The prediction (5.9 million ms) is somewhat higher than the average milliseconds seen in large-result cohorts (~5.65 million ms) but consistent with constructors who have fewer scored points in constructor_results.
- The model reliance mainly comes from aggregate patterns of experience (number of results) and known historical race times rather than any standout feature in this constructor's specific records.
- Since no individual cell value has a strong influence score, the prediction is stable and reflects general trends from constructors with similar race participation and performance rather than any unique exceptional or underperforming events.

Thus, the prediction tells us that this constructor's average race times align with typical values for similarly active participants with low-scoring finishes, showing no unusual outlier performances in the data influencing the model's output.

⏏ MODEL PERFORMANCE

254396.094

MAE

67328999424.000

MSE

259478.328

RMSE